

A First Look at Your Project's Size

CS 5388 / CS 4390

Team: 10 Project: DineCheck (Campus Dining System)

Work through every step together as a team. Have your Sprint 1 charter open (scope boundary and stakeholder map) and your interview notes. Everything you write here becomes the starting point for today's user stories and for the two estimates in Part 2.

Step 1. List what Release 1 must do (as a team)

Start from the "in scope" items in your charter and the strongest needs from your interviews. Write 5 to 8 capabilities as what the system must do, not how. Then mark the hidden work each one will need using the key below.

Hidden work key: T = Testing S = Security or privacy I = Integration with other systems D = Data (collection, cleaning, migration) H = Hosting and deployment U = Usability and accessibility O = Documentation or training

#	Capability (what, not how)	Evidence (charter or interview)	Hidden work
Ex.	<i>A student can buy a semester parking permit online.</i>	<i>Interview 4, Parking Services</i>	<i>T, S, I, H</i>
1	<i>Students can view campus dining locations, hours, and whether each location is open.</i>	<i>Interview commuting sophomore</i>	<i>T, I, D, H, U</i>
2	<i>The system works on phones and provides easy to scan information.</i>	<i>Interview time pressed students</i>	<i>T, H, U</i>
3	<i>Students finding dining location sold-out items within the menu.</i>	<i>Focused availability interviews: sold-out ribs, soup, and sandwiches</i>	<i>T, I, D, H, U, O</i>
4	<i>Students can filter through results for their dietary restrictions/preferences</i>	<i>Vegetarian, allergy, and meal plan participants described these constraints.</i>	<i>T, D, U</i>
5	<i>Students can view a current crowd level and estimated wait time for each location.</i>	<i>Commuting sophomore chose the shortest line</i>	<i>T, I, D, H, U</i>

The example row is from the fictional Campus Parking System used in lecture.

Step 2. Take the outside view: your own experience (as a team)

As a team, pick the most similar thing any of you has built before: a class project, internship feature, hackathon app, or side project.

Question	Answer
What was it?	Streaming-App (Music/Video)
How long was it expected to take?	3 months
How long did it take?	3 months
What took longer than expected?	Coordinating GitHub changes, integrating team content, responsive layout testing, and deployment details.

Step 3. Take the outside view: similar products *(as a team, web search)*

Divide the searching, then decide together. Find three real products that solve a problem close to yours. Include at least one small product (a startup or an app from a small team) and at least one big product (a large company platform or a government system). For each, find out how long it took to reach its first public release, what went well, and what went wrong.

How to find the timeline and the story

1. Launch announcements, company blogs, and founder interviews often say when building started and when the product first went public.
2. Press coverage, app store version histories, and the Internet Archive's Wayback Machine help confirm first release dates.
3. Postmortems, conference talks, and, for government systems, GAO and Inspector General reports explain what worked and what did not.
4. Measure from the start of development to the first public release, and write down what that first release included. A limited first version and a full launch are different numbers.
5. Every date and figure needs a source you can link. If you cannot find one, write "unknown." AI tools may suggest products to look at, but they are not a source.

Example: a completed entry for a parking product

ParkEasy is fictional. It shows the level of detail to aim for, not real data.

Field	Entry
Product and company	ParkEasy, a four-person startup
Why it is similar	Drivers find and pay for parking from a phone, like our Campus Parking System.
Development started	January 2019 (founder blog post)
First public release	October 2019, in one city only (local news launch story)
Time to first release	About 9 months
What the first release included	A map of open lots and pay-by-phone. No reservations and no monthly permits.
Team size	Three engineers and one designer
What worked	Launching in one city first. Partnering with a single lot operator to get reliable availability data.
What did not work	Their own parking sensors gave bad availability data, so they switched to the operator's feed, adding about 3 months. Payment integration took twice as long as planned.
Sources	Founder blog post (link); local news launch story (link)

What this example tells a team building a parking app: plan for data and payment integration to take the longest, and scope the first release to one location.

Step 3 worksheet: similar products

	Product A (small)	Product B (big)	Product C (your choice)
Product and company	Waitwhile - Waitwhile	Grubhub - Grubhub	Google Maps – Google/Alphabet
Why it is similar to your project	Lets customers join a virtual line and wait elsewhere instead of standing in a crowded location.	Lets customers find restaurants and order takeout or delivery instead of waiting at a busy restaurant. It does not primarily predict crowd levels.	Helps users avoid crowded restaurants and stores using historical and live busyness data.
Development started	Not publicly verified, but 2017 is its founding year, public launch was 2016.	2004	2003
First public release	2016	2004 in Chicago.	February 8, 2005
Time to first release	Unknown: Development start date not verified.	Within 2004; exact number of months unknown.	Approximately 2 years, based on a 2003 start and February 2005 launch.
What the first release included	A virtual waitlist for managing customers' place in line	A website where users can enter an address to find restaurants that are delivered there and view delivery information and menus.	A desktop website with maps and directions. Popular Times was not part of the first release.
Team size, if known	Two co-founders are identified; total launch-team size unknown.	Two founders, Matt Maloney and Mike Evans; total launch-team size unknown.	Four people are identified with the original Where 2 Technologies team; Google's launch-team size is unknown.
What worked	Remote queuing addressed the immediate problem of standing in line; the product later attracted many business customers.	Address-based search answered a specific question: "Who delivers to me?"	Simple maps and directions gave users a useful starting point; later business information added a way to plan visits.
What did not work	Its original platform later became slow and had outages at scale, prompting a rebuild.	Ordering food avoids a restaurant visit but does not tell users how crowded a place is or helps with stores.	Business information is unavailable when Google lacks sufficient visit data; it is not an exact headcount.
Sources (links)	https://waitwhile.com/blog/meet-the-new-waitwhile/ https://www.linkedin.com/company/waitwhile https://people.equilar.com/bio/org/waitwhile/4345924	https://about.wonder.com/about-us/default.aspx https://en.wikipedia.org/wiki/Grubhub	https://blog.google/products-and-platforms/products/maps/look-back-15-years-mapping-world/ https://support.google.com/business/answer/6263531?hl=en https://en.wikipedia.org/wiki/Google_Maps

Step 4. Compare *(as a team)*

1. Look at the hidden-work column in Step 1. Which letter appears most often? Did any of your similar products struggle with that same kind of work?
 - T (Testing) and U (Usability) tied for most often; each appears in all five capabilities. Most of the products we have been involved with required through scales for testing features or workflows.
2. How long did your similar products take to reach a first release, and with how many people? What does that suggest about a first release of your project?
 - Usually a rushed 3 months, with a group of 2-3 team members. This would suggest that while feasible in time scale for this class, the first release of this project would still require an arduous amount of time hours put into it to get a semblance of first release candidate.
3. Name one practice from the “what worked” rows your team will copy, and one mistake from the “what did not work” rows you will avoid.
 - We will copy business information such as open hours and options like gluten-free so people can plan the place they would want to visit accordingly and not waste their time. A mistake we will avoid is thinking too small about the scale, keeping us from having to rebuild and having outages.

Our comparables suggest a first release would take _2_ to _3_ months with a team of _4_.

Keep this sheet. You will use Step 1 to write user stories, and Steps 2 and 3 give you comparators for analogous estimation in Part 2.